# The Continuity Equations

Yen-Ting Wu (Lecture Note)

## The Continuity Equations

**All the equations we will discuss in this note:**

$$\begin{cases} \text{Continuity for Charge} : \dfrac{\partial \varrho(\mathbf{r},t)}{\partial t} + \boldsymbol{\nabla} \cdot \mathbf{J} = 0 \\ \text{Continuity for Energy} : \dfrac{\partial u_{\text{em}}(\mathbf{r},t)}{\partial t} + \boldsymbol{\nabla} \cdot \mathbf{S} = 0 \\ \text{Continuity for Momentum} : \dfrac{\partial \boldsymbol{g}}{\partial t} - \boldsymbol{\nabla} \cdot \overleftrightarrow{\mathbf{T}} = 0 \end{cases}$$

where $u_{\text{em}}$ is the total energy of the electric field and magnetic field, $\mathbf{S}$ is the **Poynting** vector, $\mathbf{g}$ is the momentum density stored in the field and $\overleftrightarrow{\mathbf{T}}$ is the **Maxwell's Stress Tensor**.

## 1 Continuity Equation for the Charge

We have the current density $\mathbf{J}(\mathbf{r},t)$, then the current is denoted by

$$I = \int \mathbf{J} \cdot d\mathbf{a} = \oint_{\partial \mathcal{V}} \mathbf{J} \cdot d\mathbf{a}. \tag{1}$$

The first integral means that the current density $\mathbf{J}$ which is defined as

$$\mathbf{J} = \frac{d\mathbf{I}}{da_{\perp}}, \tag{2}$$

through a surface perpendicular to it represents the electric current through that surface at that instant. It describes the rate at which electric charge passes through the surface per unit time; and the second integral means that we choose a region that has a boundary(or surface), and we discuss the rate at which electric charge passes through the boundary of the region. Apply the divergence theorem, we obtain

$$\oint_{\partial \mathcal{V}} \mathbf{J} \cdot d\mathbf{a} = \int_{\mathcal{V}} \left( \boldsymbol{\nabla} \cdot \mathbf{J} \right) d\tau. \tag{3}$$

Note that the divergence theorem gives the geometric representation that the flow is pointing "outside", in other words, the infinitesimal area vector always points outward in the bounded region. That is, there are some charges decrease (the flow points out) or increase (the flow points in) in the region as we don't consider the charges outside or appear suddenly. In other words, the tendency of charge variation is opposite to the direction of the current flux, which is why a negative sign appears, this fact leads to:

$$\int_{\mathcal{V}} \left[\boldsymbol{\nabla} \cdot \mathbf{J}(\mathbf{r}, t)\right] d\tau - \frac{dq}{dt} = -\frac{d}{dt} \int_{\mathcal{V}} \varrho(\mathbf{r}, t)\, d\tau = \int_{\mathcal{V}} \left(-\frac{\partial \varrho(\mathbf{r}, t)}{\partial t}\right) d\tau, \tag{4}$$

it yields to

$$\boxed{\frac{\partial \varrho(\mathbf{r}, t)}{\partial t} + \boldsymbol{\nabla} \cdot \mathbf{J}(\mathbf{r}, t) = 0.} \tag{5}$$

(5) is a local equation in whatever coordinate the density lives in, that is, this is a *local* conservation law. Note that we don't need to emphasize the source coordinates or the observation coordinates. The continuity equation is a local conservation law defined in the same coordinate space where the charge density and current density are distributed. Using $\mathbf{r}$ or $\mathbf{r}'$ is merely a change of dummy variables. The equation itself does not distinguish between source and observation coordinates, since *it describes the intrinsic time evolution of charge within the source distribution.* Sure, we could also say:

$$\int_{\mathcal{V}} \left[\boldsymbol{\nabla}' \cdot \mathbf{J}(\mathbf{r}', t)\right] d\tau' = \int_{\mathcal{V}} \left(-\frac{\partial \varrho(\mathbf{r}', t)}{\partial t}\right) d\tau',$$

it doesn't matter. The only time we need to distinguish between two coordinate systems is when we simultaneously consider the effect of a source on the field at an external observation point. We can also replace $\varrho$ by the electric field with respect of time by Gauss's Law, that is,

$$\frac{\partial\big(\epsilon_0 \boldsymbol{\nabla} \cdot \mathbf{E}\big)}{\partial t} + \boldsymbol{\nabla} \cdot \mathbf{J} = 0 \Rightarrow \boldsymbol{\nabla} \cdot \left(\epsilon_0 \frac{\partial \mathbf{E}}{\partial t} + \mathbf{J}\right) = 0 \tag{6}$$

# 2 Continuity Equation for the Energy

## 2.1 Poynting Theorem

The electromagnetic force $\mathbf{F}_{\text{em}}$ is that

$$\mathbf{F}_{\text{em}} = q\big[\mathbf{E}(\mathbf{r}, t) + \mathbf{v}(\mathbf{r}, t) \times \mathbf{B}(\mathbf{r}, t)\big],$$

the power of the work($W$) done by the electric and magnetic fields on the charges is given by

$$\frac{dW}{dt} = \mathbf{F}_{\text{em}} \cdot \mathbf{v} = q\big[\mathbf{E}(\mathbf{r}, t) + \mathbf{v}(\mathbf{r}, t) \times \mathbf{B}(\mathbf{r}, t)\big] \cdot \mathbf{v} = q\mathbf{E} \cdot \mathbf{v} = \mathbf{E} \cdot (q\mathbf{v}). \tag{7}$$

With the continuous distribution volume, the rate of the work done by the fields in the region $\mathcal{V}$ is denoted by:

$$\frac{dW}{dt} = \int_{\mathcal{V}} \mathbf{E}(\mathbf{r},t) \cdot \varrho(\mathbf{r},t)\mathbf{v}\, d\tau = \int_{\mathcal{V}} \mathbf{E}(\mathbf{r},t) \cdot \mathbf{J}(\mathbf{r},t)\, d\tau, \tag{8}$$

where $\mathbf{J} = \varrho\mathbf{v}$. Note that we are discussing the effect acted by the fields (not the source), so we use *Observation coordinates*($\mathbf{r}$) here instead of the source coordinates($\mathbf{r}'$). By the Maxwell's equations, we can eliminate the current density $\mathbf{J}$ by

$$\mathbf{J} = \frac{1}{\mu_0}\boldsymbol{\nabla} \times \mathbf{B}(\mathbf{r},t) - \epsilon_0 \frac{\partial \mathbf{E}(\mathbf{r},t)}{\partial t}. \tag{9}$$

Substituting (9) into (8), the rate of the work can be rewritten as

$$\frac{dW}{dt} = \int_{\mathcal{V}} \mathbf{E} \cdot \left( \frac{1}{\mu_0}\boldsymbol{\nabla} \times \mathbf{B} - \epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) d\tau. \tag{10}$$

The first term of the integral is

$$\begin{aligned}
\frac{1}{\mu_0}\mathbf{E} \cdot (\boldsymbol{\nabla} \times \mathbf{B}) &= \frac{1}{\mu_0} \Bigg[ \underbrace{-\boldsymbol{\nabla} \cdot \big(\mathbf{E} \times \mathbf{B}\big) + \mathbf{B} \cdot \big(\boldsymbol{\nabla} \times \mathbf{E}\big)}_{\text{Product Rules (iii)}} \Bigg] \\
&= \frac{-1}{\mu_0}\boldsymbol{\nabla} \cdot \big(\mathbf{E} \times \mathbf{B}\big) + \frac{1}{\mu_0}\mathbf{B} \cdot \left( - \underbrace{\frac{\partial \mathbf{B}(\mathbf{r},t)}{\partial t}}_{\text{Faraday's Law}} \right) \\
&= \frac{-1}{\mu_0}\boldsymbol{\nabla} \cdot \big(\mathbf{E} \times \mathbf{B}\big) - \frac{1}{\mu_0}\left( \mathbf{B} \cdot \frac{\partial \mathbf{B}}{\partial t} \right) = -\boldsymbol{\nabla} \cdot \left( \frac{1}{\mu_0}\mathbf{E} \times \mathbf{B} \right) - \underbrace{\frac{1}{\mu_0}\frac{\partial}{\partial t}\left(\frac{1}{2}B^2\right)}_{= \frac{\partial}{\partial t}\left(\frac{1}{2\mu_0}B^2\right)},
\end{aligned}$$

and the second term of the integral is

$$\mathbf{E} \cdot \left( -\epsilon_0 \frac{\partial \mathbf{E}}{\partial t} \right) = -\epsilon_0 \left( \mathbf{E} \cdot \frac{\partial \mathbf{E}}{\partial t} \right) = -\frac{\partial}{\partial t}\left(\frac{\epsilon_0}{2}E^2\right).$$

Combining the results of each term and substituting into (10), then (10) becomes

$$\begin{aligned}
\frac{dW}{dt} &= -\int_{\mathcal{V}} \boldsymbol{\nabla} \cdot \left( \frac{1}{\mu_0}\mathbf{E} \times \mathbf{B} \right) d\tau - \int_{\mathcal{V}} \frac{\partial}{\partial t}\left( \frac{\epsilon_0}{2}E^2 + \frac{1}{2\mu_0}B^2 \right) d\tau \\
&= -\oint_{\partial\mathcal{V}} \frac{1}{\mu_0}\mathbf{E} \times \mathbf{B} \cdot d\mathbf{a} - \int_{\mathcal{V}} \frac{\partial u_{\text{em}}}{\partial t} d\tau,
\end{aligned} \tag{11}$$

where $u_{\text{em}}$ is the electromagnetic energy density, which is the sum of the energy density of electric field and the energy density of the magnetic field, i.e.

$$u_{\text{em}} = u_{\text{e}} + u_{\text{m}} = \frac{\epsilon_0}{2}E^2 + \frac{1}{2\mu_0}B^2. \tag{12}$$

In (11), we can further define a new quantity $\mathbf{S}$, which can be denoted by

$$\mathbf{S} \coloneqq \frac{1}{\mu_0}\mathbf{E} \times \mathbf{B}. \tag{13}$$

$\mathbf{S}$ is called **Poynting Vector**. This physical quantity describes the orientation of the *energy flow* from the fields.

## 2.2 Continuity and Conservation

By the definition of Poynting vector $\mathbf{S}$ in (13), we substitute it into (11), the rate of the work is rewritten as:

$$\frac{dW_{\text{Field to Charges}}}{dt} = -\int_{\mathcal{V}} \boldsymbol{\nabla} \cdot \mathbf{S}\, d\tau - \int_{\mathcal{V}} \frac{\partial u_{\text{em}}}{\partial t} d\tau = -\oint_{\mathcal{S}} \mathbf{S} \cdot d\mathbf{a} - \int_{\mathcal{V}} \frac{\partial u_{\text{em}}}{\partial t} d\tau \tag{14}$$

The negative sign indicates that when the electric and magnetic fields perform work on charges, **the energy of the fields decreases, transferring energy to the charges.** Moreover, through the Poynting vector, this equation specifies the **directional flow of energy**. In this way, the equation provides a perspective on *energy conservation* in the presence of electromagnetic fields.
When the rate of the work done by electromagnetic field is zero with the time, that is, $dW/dt = 0$. Then (14) becomes:

$$-\int_{\mathcal{V}} \boldsymbol{\nabla} \cdot \mathbf{S}\, d\tau - \int_{\mathcal{V}} \frac{\partial u_{\text{em}}}{\partial t}\, d\tau = 0 \Rightarrow \int_{\mathcal{V}} \boldsymbol{\nabla} \cdot \mathbf{S}\, d\tau + \int_{\mathcal{V}} \frac{\partial u_{\text{em}}}{\partial t}\, d\tau = 0,$$

and this yields to the final equation, the **continuity equation for energy**:

$$\boxed{\frac{\partial u_{\text{em}}}{\partial t} + \boldsymbol{\nabla} \cdot \mathbf{S} = 0.} \tag{15}$$

Note that the conservation in this note is *local*. (You cannot expect energy or charge to suddenly appear in some corner of the universe outside the scope of our discussion just to satisfy our local conservation.)
But somehow, $\mathbf{S}$ and the energy are *NOT* unique, instead, they are "infinitely many"! We may have another possible answers or terms which can be added to the equation but still remain the conservation law hold. For instance,

$$\begin{cases} u'_{\text{em}} = \dfrac{\epsilon_0}{2}E^2 + \dfrac{1}{2\mu_0}B^2 + \dfrac{\star}{n}W_0^2 \Rightarrow \dfrac{\partial u'_{\text{em}}}{\partial t} = \dfrac{\partial u_{\text{em}}}{\partial t} \\ \mathbf{S}' = \mathbf{S} + \boldsymbol{\nabla} \times \mathbf{W} \Rightarrow \boldsymbol{\nabla} \cdot \mathbf{S}' = \boldsymbol{\nabla} \cdot \mathbf{S}, \end{cases}$$

in this case, conservation law still holds. So, in mathematics, the conserved charges[1] have infinite possibilities, but in physics, we choose the gauge that the quantities satisfying the conservation law have the property of symmetry under the rotation or coordinates change.

# 3 Continuity Equation for the Momentum

## 3.1 Maxwell's Stress Tensor

The electromagnetic force perm unit volume is

$$\mathbf{f} = \varrho\mathbf{E} + \mathbf{J} \times \mathbf{B} \tag{16}$$

by Lorenz's force law. It can be written in the form of $\mathbf{E}$ and $\mathbf{B}$ that is:

$$\begin{aligned}\mathbf{f} &= \epsilon_0\big(\boldsymbol{\nabla} \cdot \mathbf{E}\big)\mathbf{E} + \left(\frac{1}{\mu_0}\boldsymbol{\nabla} \times \mathbf{B} - \epsilon_0\frac{\partial \mathbf{E}}{\partial t}\right) \times \mathbf{B} \\ &= \epsilon_0\big(\boldsymbol{\nabla} \cdot \mathbf{E}\big)\mathbf{E} + \frac{1}{\mu_0}\big(\boldsymbol{\nabla} \times \mathbf{B}\big) \times \mathbf{B} - \epsilon_0\frac{\partial \mathbf{E}}{\partial t} \times \mathbf{B},\end{aligned} \tag{17}$$

In the equation, we have

$$\begin{aligned}-\epsilon_0\frac{\partial \mathbf{E}}{\partial t} \times \mathbf{B} &= -\epsilon_0\frac{\partial}{\partial t}(\mathbf{E} \times \mathbf{B}) + \underbrace{\epsilon_0\mathbf{E} \times \frac{\partial \mathbf{B}}{\partial t}}_{\text{Faraday's Law}} \\ &= -\epsilon_0\mu_0\frac{\partial}{\partial t}\left(\frac{1}{\mu_0}\mathbf{E} \times \mathbf{B}\right) + \epsilon_0\mathbf{E} \times (-\boldsymbol{\nabla} \times \mathbf{E}),\end{aligned} \tag{18}$$

where we have

$$-\epsilon_0\mu_0\frac{\partial}{\partial t}\left(\frac{1}{\mu_0}\mathbf{E} \times \mathbf{B}\right) = -\epsilon_0\mu_0\frac{\partial \mathbf{S}}{\partial t}. \tag{19}$$

Now, we have two similar terms, which are $(\boldsymbol{\nabla} \times \mathbf{B}) \times \mathbf{B}$ and $\mathbf{E} \times (\boldsymbol{\nabla} \times \mathbf{E})$, by Products Rules (ii), we can obtain

$$\begin{cases}\boldsymbol{\nabla}\big(E^2\big) = 2\mathbf{E} \times (\boldsymbol{\nabla} \times \mathbf{E}) + 2(\mathbf{E} \cdot \boldsymbol{\nabla})\mathbf{E} \\ \boldsymbol{\nabla}\big(B^2\big) = 2\mathbf{B} \times (\boldsymbol{\nabla} \times \mathbf{B}) + 2(\mathbf{B} \cdot \boldsymbol{\nabla})\mathbf{B},\end{cases}$$

[1]"Charges" here are not only the electric charges, instead, we may call some quantities or scalars charges which satisfy the conservation law.

therefore:

$$\begin{cases} -\epsilon_0 \mathbf{E} \times (\boldsymbol{\nabla} \times \mathbf{E}) = \epsilon_0 (\mathbf{E} \cdot \boldsymbol{\nabla})\mathbf{E} - \dfrac{\epsilon_0}{2} \boldsymbol{\nabla}\left(E^2\right) \\ \dfrac{1}{\mu_0}(\boldsymbol{\nabla} \times \mathbf{B}) \times \mathbf{B} = \dfrac{1}{\mu_0}(\mathbf{B} \cdot \boldsymbol{\nabla})\mathbf{B} - \dfrac{1}{2\mu_0} \boldsymbol{\nabla}\left(B^2\right). \end{cases} \tag{20}$$

Finally, we substitute (20), (19) and (18) into (17), the electromagnetic force perm unit volume $\mathbf{f}$ is now

$$\begin{aligned} \mathbf{f} = \epsilon_0 &\left[ (\boldsymbol{\nabla} \cdot \mathbf{E})\mathbf{E} + (\mathbf{E} \cdot \boldsymbol{\nabla})\mathbf{E} - \frac{1}{2} \boldsymbol{\nabla}(E^2) \right] \\ &+ \frac{1}{\mu_0} \left[ (\boldsymbol{\nabla} \cdot \mathbf{B})\mathbf{B} + (\mathbf{B} \cdot \boldsymbol{\nabla})\mathbf{B} - \frac{1}{2} \boldsymbol{\nabla}(B^2) \right] - \epsilon_0 \mu_0 \frac{\partial \mathbf{S}}{\partial t}. \end{aligned} \tag{21}$$

We can find a really astonishing symmetry of the equation, but this seems like a dog shit, so we can rewrite this in the form of a tensor:

$$\begin{aligned} \left( \mathbf{f} + \epsilon_0 \mu_0 \frac{\partial \mathbf{S}}{\partial t} \right)_i &= \epsilon_0 \left[ \partial_j E_j E_i + E_j \partial_j E_i - \frac{1}{2} \partial_i E^2 \right] + \frac{1}{\mu_0} \left[ \partial_j B_j B_i + B_j \partial_j B_i - \frac{1}{2} \partial_i B^2 \right] \\ &= \epsilon_0 \left[ \partial_j (E_i E_j) - \underbrace{\partial_j \left( \frac{1}{2} \delta_{ij} E^2 \right)}_{\text{To make } \partial_j \text{ be } \partial_i} \right] + \frac{1}{\mu_0} \left[ \partial_j (B_i B_j) - \partial_j \left( \frac{1}{2} \delta_{ij} B^2 \right) \right]. \end{aligned}$$

As every terms have **two** indies and a $\partial_j$, then we can "predict" that there's a *Rank-2* tensor and a divergence to express this shit. This Tensor is called **Maxwell's Stress Tensor** $\overleftrightarrow{\mathbf{T}}$, which is denoted by

$$T_{ij} = \epsilon_0 \left( E_i E_j - \frac{1}{2} \delta_{ij} E^2 \right) + \frac{1}{\mu_0} \left( B_i B_j - \frac{1}{2} \delta_{ij} B^2 \right), \tag{22}$$

and therefore we can check:

$$\left( \boldsymbol{\nabla} \cdot \overleftrightarrow{\mathbf{T}} \right)_i = \partial_j T_{ij} = \partial_j [\cdots] \quad \checkmark \tag{23}$$

So, the stress tensor is a rank-2 tensor while the divergence of the tensor is a vector point to the $i$-th direction. Then the force per unit volume becomes

$$\mathbf{f} = \boldsymbol{\nabla} \cdot \overleftrightarrow{\mathbf{T}} - \epsilon_0 \mu_0 \frac{\partial \mathbf{S}}{\partial t}. \tag{24}$$

The total electromagnetic force on the charges in volume $\mathcal{V}$ is given by

$$\mathbf{F} = \int_{\mathcal{V}} \mathbf{f}\, d\tau = \int_{\mathcal{V}} \boldsymbol{\nabla} \cdot \overleftrightarrow{\mathbf{T}}\, d\tau - \epsilon_0 \mu_0 \int_{\mathcal{V}} \frac{\partial \mathbf{S}}{\partial t}\, d\tau = \oint_{\partial \mathcal{V}} \overleftrightarrow{\mathbf{T}} \cdot d\mathbf{a} - \epsilon_0 \mu_0 \frac{d}{dt} \int_{\mathcal{V}} \mathbf{S}\, d\tau. \tag{25}$$

The stress tensor $T_{ij}$ describes the transport of momentum in a continuous medium or physical field. Specifically, the component $T_{ij}$ represents the flux of the $i$-component of momentum through a surface whose normal direction is along the $j$ axis. In other words, it quantifies how momentum in a given direction flows across a surface in space. Under the static situation, which means that $\mathbf{S}$ is a constant (the energy flux remains the same direction and amplitude), the Force on the charges is

$$\begin{cases} \mathbf{F} = \oint_{\partial\mathcal{V}} \overleftrightarrow{\mathbf{T}} \cdot d\mathbf{a} \\ F_i = \oint_{\partial\mathcal{V}} T_{ij}\hat{n}_j \, da \quad \text{For } i\text{-component} \end{cases} \tag{26}$$

## 3.2 Continuity and Conservation

By (25), we can define a new physical quantity: the **momentum density** (momentum per unit volume) in the fields

$$\boldsymbol{g} := \mu_0\epsilon_0\mathbf{S} = \epsilon_0(\mathbf{E}\times\mathbf{B}), \tag{27}$$

so the momentum stored in the fields is

$$\mathbf{P} = \int_{\mathcal{V}} \boldsymbol{g}\, d\tau = \mu_0\epsilon_0 \int_{\mathcal{V}} \mathbf{S}\, d\tau. \tag{28}$$

In classical mechanics, we've learned that the force equals the rate of change of momentum. Then the force acting on the charges by the fields can be denoted by

$$\mathbf{F} = \frac{d\mathbf{P}_{\text{mech}}}{dt} \tag{29}$$

where $\mathbf{P}_{\text{mech}}$ is the mechanical momentum of the charges in the region $\mathcal{V}$. Hence, (25) is now be rewritten as

$$\frac{d\mathbf{P}_{\text{mech}}}{dt} = \oint_{\partial\mathcal{V}} \overleftrightarrow{\mathbf{T}} \cdot d\mathbf{a} - \frac{d}{dt}\int_{\mathcal{V}} \boldsymbol{g}\, d\tau = \int_{\mathcal{V}} \boldsymbol{\nabla}\cdot\overleftrightarrow{\mathbf{T}}\, d\tau - \int_{\mathcal{V}} \frac{\partial \boldsymbol{g}}{\partial t}\, d\tau. \tag{30}$$

This equation provide some perspectives and information: when the mechanical momentum of the charges increases,

1. The momentum stored in the fields decreases to transfer the momentum to the charges,

2. The fields carry momentum to the charges in the volume $\mathcal{V}$ through the surface.

One of the options holds, or maybe both. But if the mechanical momentum of the charges in the region is not changing, but the fields do "act" on the charges in $\mathcal{V}$, then both of the options hold and satisfy

$$\oint_{\partial\mathcal{V}} \overleftrightarrow{\mathbf{T}} \cdot d\mathbf{a} - \int_{\mathcal{V}} \frac{\partial \boldsymbol{g}}{\partial t}\, d\tau = 0,$$

hence, another continuity equation appears:

$$\boxed{\frac{\partial \boldsymbol{g}}{\partial t} - \boldsymbol{\nabla} \cdot \overleftrightarrow{\mathbf{T}} = 0} \tag{31}$$

# Some Identities

**Triple Products:**

$$\begin{cases} \mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) = \mathbf{B} \cdot (\mathbf{C} \times \mathbf{A}) = \mathbf{C} \cdot (\mathbf{A} \times \mathbf{B}) \\ \mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B}) \quad \text{Bac(k)-Cab law} \end{cases}$$

**Products Rules:**

$$\begin{cases} \boldsymbol{\nabla}(fg) = f(\boldsymbol{\nabla} g) + g(\boldsymbol{\nabla} f) \\ \boldsymbol{\nabla}(\mathbf{A} \cdot \mathbf{B}) = \mathbf{A} \times (\boldsymbol{\nabla} \times \mathbf{B}) + \mathbf{B} \times (\boldsymbol{\nabla} \times \mathbf{A}) + (\mathbf{A} \cdot \boldsymbol{\nabla})\mathbf{B} + (\mathbf{B} \cdot \boldsymbol{\nabla})\mathbf{A} \\ \boldsymbol{\nabla} \cdot (\mathbf{A} \times \mathbf{B}) = \mathbf{B} \cdot (\boldsymbol{\nabla} \times \mathbf{A}) - \mathbf{A} \cdot (\boldsymbol{\nabla} \times \mathbf{B}) \\ \boldsymbol{\nabla} \cdot (f\mathbf{A}) = f(\boldsymbol{\nabla} \cdot \mathbf{A}) + \mathbf{A} \cdot \boldsymbol{\nabla} f \\ \boldsymbol{\nabla} \times (f\mathbf{A}) = f(\boldsymbol{\nabla} \times \mathbf{A}) - \mathbf{A} \times \boldsymbol{\nabla} f \\ \boldsymbol{\nabla} \times (\mathbf{A} \times \mathbf{B}) = (\mathbf{B} \cdot \boldsymbol{\nabla})\mathbf{A} - (\mathbf{A} \cdot \boldsymbol{\nabla})\mathbf{B} + \mathbf{A}(\boldsymbol{\nabla} \cdot \mathbf{B}) - \mathbf{B}(\boldsymbol{\nabla} \cdot \mathbf{A}) \end{cases}$$

**Second Derivatives:**

$$\begin{cases} \boldsymbol{\nabla} \cdot (\boldsymbol{\nabla} \times \mathbf{A}) = 0 \\ \boldsymbol{\nabla} \times (\boldsymbol{\nabla} f) = \mathbf{0} \\ \boldsymbol{\nabla} \times (\boldsymbol{\nabla} \times \mathbf{A}) = \boldsymbol{\nabla}(\boldsymbol{\nabla} \cdot \mathbf{A}) - \nabla^2 \mathbf{A} \end{cases}$$

**Green's Identities and Some Integrals:**

$$\begin{cases} \displaystyle\int_{\mathcal{V}} \left(T\,\nabla^2 U - U\nabla^2 T\right) d\tau = \oint_{\mathcal{S}} \left(T\boldsymbol{\nabla} U - U\boldsymbol{\nabla} T\right)\cdot d\mathbf{a} \\ \displaystyle\int_{\mathcal{S}} \boldsymbol{\nabla} T \times d\mathbf{a} = -\oint_{\mathcal{P}} T\,d\boldsymbol{\ell} \\ \displaystyle\oint (\mathbf{c}\cdot\mathbf{r})\,d\boldsymbol{\ell} = \left(\int d\mathbf{a}\right)\times\mathbf{c} \\ \displaystyle\int_{\mathcal{V}} \boldsymbol{\nabla}\times\mathbf{F}\,d\tau = -\oint_{\mathcal{S}} \mathbf{F}\times d\mathbf{a} \end{cases}$$

**Delta functions:**

$$\begin{cases} \displaystyle\boldsymbol{\nabla}\cdot\left(\frac{\hat{\boldsymbol{\mathscr{r}}}}{\mathscr{r}^2}\right) = 4\pi\delta^3(\boldsymbol{\mathscr{r}}) \\ \displaystyle\boldsymbol{\nabla}\left(\frac{1}{\mathscr{r}}\right) = \boldsymbol{\nabla}\left(\frac{1}{\mathbf{r}-\mathbf{r}'}\right) = -\frac{\hat{\boldsymbol{\mathscr{r}}}}{\mathscr{r}^2} \\ \displaystyle\boldsymbol{\nabla}'\left(\frac{1}{\mathscr{r}}\right) = \boldsymbol{\nabla}'\left(\frac{1}{\mathbf{r}-\mathbf{r}'}\right) = \frac{\hat{\boldsymbol{\mathscr{r}}}}{\mathscr{r}^2} \\ \displaystyle\nabla^2\left(\frac{1}{\mathscr{r}}\right) = -4\pi\delta^3(\boldsymbol{\mathscr{r}}) \end{cases}$$